CineCode: Movie Explorer Project Proposal

Team Information

Item	Details
Team name	CineCode
App title	CineCode: Movie Explorer
Student 1	Yaa Adarkwaa
Student 2 (optional)	Kuku Abdulquadri
GitHub repository	https://github.com/frimpomaahadarkwahyaa/CineCode-Movie-Explorer

Project Overview

CineCode is a React Native mobile app that lets movie fans search for films, browse live results, and read detailed information, all pulled from a third-party API with no local database.

Objectives

- Let users find movies quickly by keyword and see what is trending.
- Present key information (poster, title, rating, release year) in a clean, scrollable list.
- Give each movie a details screen with overview, genres, runtime and rating.
- Let users save favorite movies for later during a session.
- Meet the course requirements: Axios with async/await, React Hooks, React Navigation, FlatList, reusable components, Flexbox, and Expo vector icons.

Target users

- Casual movie watchers deciding what to watch tonight.
- Students and film fans looking up ratings, release dates and plot summaries.
- Anyone who wants a fast, simple alternative to browsing a full movie website on a phone.

Third-Party API: The Movie Database (TMDB)

The app will use the free TMDB API, which returns live movie data and image paths for posters and backdrops. A free API key is required and will be stored in a config file kept out of the repository.

Purpose	Endpoint
Trending movies (Home)	/trending/movie/week
Search by keyword	/search/movie?query=...
Movie details	/movie/{movie_id}

All requests are made with Axios using async/await. Poster images load from the TMDB image base URL.

Main Features and User Stories

Features

- Search movies by keyword.
- Trending movies shown on the home screen.
- Scrollable results list with poster, title, year and rating.
- Movie details screen with overview, genres, runtime and rating.
- Favorites list (add and remove).
- Loading spinner and friendly error messages when the network fails.
- Layout that adapts to different screen sizes.

User stories

- As a user, I want to search for movies using keywords.
- As a user, I want to browse a list of movies with key information.
- As a user, I want to tap a movie to view detailed information.
- As a user, I want to save movies I like to a favorites list.
- As a user, I want a layout that works on various screen sizes.

Planned Screens and Navigation

The app has four screens. A bottom tab navigator holds Home and Favorites, and a stack navigator opens Results and Details on top of Home.

Screen	Description	Navigates to
Home / Search	Search bar, trending movies carousel, and a button to search. Entry point of the app.	Results, Details
Results List	FlatList of movies matching the search, each row showing poster, title, year and rating.	Details
Movie Details	Large poster, title, rating, release date, runtime, genres, overview, and an add-to-favorites button.	Back to Results / Home
Favorites (extra)	FlatList of saved movies with a remove option.	Details

Technical Implementation Plan

Requirement	How CineCode will use it
Axios	All TMDB calls go through one Axios instance (base URL plus API key) using async/await with try/catch.
React Hooks	useState for search text, results and favorites; useEffect to load trending movies and fetch details on screen open; useReducer for the favorites list (add/remove).
Custom hook	useApiData() wraps fetching and returns data, loading and error so screens stay free of API logic.
React Navigation	Bottom tabs (Home, Favorites) plus a native stack for Results and Details, passing the movie id as a route parameter.
FlatList	Used for the results list, trending row and favorites, with keyExtractor and lightweight row components for efficient rendering.

Requirement	How CineCode will use it
Reusable components	SearchBar, MovieCard, ListItem, Header, LoadingSpinner and ErrorMessage, shared across screens.
Flexbox and Box Model	StyleSheet layouts using flex, flexDirection, alignItems, padding, margin and position for responsive screens.
Icons and images	Ionicons from Expo Vector Icons (search, heart, star, back) and TMDB poster images.

Planned file structure: components/, screens/, hooks/, services/ (Axios setup), navigation/, and App.js.

Wireframes and GitHub Repository

Three wireframes are included with this proposal, one per core screen, showing the layout and the navigation between them.

1. Wireframe 1: Home / Search screen
2. Wireframe 2: Results List screen
3. Wireframe 3: Movie Details screen
4. Wireframe 4 (optional): Favorites screen

The GitHub repository will contain this proposal document, the wireframe images, the starter project source code with meaningful commits, and a README.md covering the project overview, setup steps, API used, team contributions, and screenshots.

Team Responsibilities and Timeline

Individual responsibilities

Member	Responsibilities
Yaa Adarkwaa	React Navigation setup, Home and Details screens, wireframes, README and GitHub repo
Kuku Abdulquadri	Axios API service, useApiData hook, Results list, Favorites screen, reusable components

All members will commit to the shared GitHub repository and test the app together.

Development timeline

Week	Milestone
Week 1	Proposal, wireframes, GitHub repo, TMDB API key, Expo project setup
Week 2	Axios service, useApiData hook, navigation, Home and Results screens with FlatList
Week 3	Details and Favorites screens, reusable components, loading and error handling
Week 4	Styling and responsiveness, testing, README and screenshots, video presentation, final submission